
Generation of φ^4 Feynman Graphs Through a Recursive Algorithm

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Abstract

Generating all possible Feynman diagrams is crucial in quantum field theory for computing interaction amplitudes. In scalar φ^4 theory, characterized by four-point interaction vertices, these diagrams can be translated into mathematical graphs, enabling a systematic graph-theoretic approach. This project aimed to develop an open-source program capable of generating and counting all φ^4 Feynman graphs with up to nine points of interaction. While previous work has generated graphs with few interaction vertices, this algorithm was designed to efficiently scale to more complex cases, and thus generate larger graphs. A family of graphs, known as vacuum graphs, was recursively generated and then systematically modified to create all valid φ^4 Feynman graphs with up to nine points of interaction. Each major component of the algorithm was timed individually, and the number of generated graphs was recorded. The algorithm successfully generated all φ^4 Feynman graphs with up to nine vertices. Preliminary results indicate that the most time-consuming part of the algorithm is the isomorphism check, and that the number of generated graphs grows combinatorially. Although the program currently performs efficiently, scalability could be improved by integrating more optimized isomorphism checks. The generated graphs are essential in particle physics, offering insight into particle interactions and potential new particles, but also relevant in statistical mechanics. Further development, such as implementing new operators to produce other kinds of Feynman graphs, would broaden the program's applicability even more.

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1 Introduction

In the mid-nineteenth century, the concept of fields was introduced by Michael Faraday. Compared to Newtonian physics, which describes the fundamental laws of physics through forces between pairs of particles, field theory offers advantages when considering special relativity. This is because special relativity imposes a speed limit that Newtonian physics does not. Many scientific discoveries later, quantum field theory was developed. This discipline, combining field theory with special relativity and quantum mechanics, formulates theories of the electroweak and strong forces that construct the Standard Model. [1]

1.1 Quantum Field Theory

Quantum field theory includes quantum chromodynamics, describing how protons and neutrons behave in atomic nuclei, quantum electrodynamics, explaining how electrons and photons interact, and other field interactions [2]. In quantum field theory, particles are viewed as perturbations of underlying fields, and their scatterings as field interactions [3]. Mathematically, these fields are defined by field operators that create and annihilate particles in space-time [4]. An interaction involving n particles can be modeled using φ^n theory [5]. Studying particle scatterings provides insight not only into the interactions themselves, but also into other systems through statistical mechanics [6].

1.2 Statistical Mechanics

Statistical mechanics addresses the behavior of macroscopic systems using probabilistic models. A key concept in this branch of study is universality, referring to the phenomenon where certain macroscopic properties are identical across distinct systems, despite differences in their microscopic structures [6].

This can be explained using renormalization group theory, a framework suggesting that as objects are observed on a larger scale, fine details become less significant [6]. An important aspect of universality is critical exponents, defined by the power-law behavior of physical quantities near phase-transitions. For instance, the Ising model, described in

subsection 1.2.1, which can be simplified to

$$M \propto (T - T_C)^\beta \quad (1)$$

has the critical exponent β . In the equation, M is the magnetization, T the temperature and T_C the critical temperature. The symbol $\propto$ denotes proportionality, and indicates, in this case, that M is proportional to $(T - T_C)^\beta$ [7].

Systems with the same critical exponent, and consequently exhibit similar characteristics, are said to be in the same universality class. Studying the critical exponent of a system within a universality class provides insight into all systems in that class. This allows the use of simple models to understand complex phenomena [7]. For example, φ^4 theory can be used to study the classical magnetism modeled by the Ising model [8].

1.2.1 The Ising model and φ^4 Theory

The Ising model describes classical magnetism in a lattice consisting of the magnetic moments S_i and S_j , which can take the values ± 1 . It is described mathematically by the Hamiltonian

$$H(\sigma) = \sum_{ij} S_i C_{ij} S_j - h \sum_i S_i \quad (2)$$

defining the energy of a given configuration σ . In the equation, C_{ij} is a matrix describing the interaction between S_i and S_j , while h is an external magnetic field acting on the lattice. The sums are taken over the sites of a d -dimensional lattice, and for $d \geq 3$ the model is difficult to solve numerically. Therefore, simpler systems in the same universality class, such as φ^4 theory, are used to study the Ising model more easily. [9]

The behavior of fields varying over time and space is described by φ^4 theory, mathematically defined by the Lagrangian

$$\mathcal{L}(\varphi) = \frac{1}{2}(\partial_\mu \varphi)(\partial^\mu \varphi) - \frac{1}{2}m^2 \varphi^2 - \frac{\lambda}{4!} \varphi^4 \quad (3)$$

representing the energy density of a given field φ . In the equation, $\partial_\mu \varphi$ and $\partial^\mu \varphi$ are partial derivatives with respect to space and time, m is the mass of the particles associated with the field, and λ is the strength of the interaction [10].

1.3 Particle Physics

Particle physics involves the study of subatomic particles: matter, antimatter and particles mediating the fundamental interactions, as outlined by quantum field theory. These are considered point-like and have properties, such as mass and spin [11]. Examples of subatomic particles are fermions and bosons [12].

Fermions are subatomic particles characterized by a spin equal to an odd multiple of $\frac{1}{2}$. This category include leptons, such as electrons and muons, and baryons, which include protons and neutrons [13]. Bosons are characterized by an integer spin, and include, among other particles, photons and gluons [14].

1.4 Feynman Diagrams

Feynman diagrams, established by Richard Feynman in the mid-twentieth century, show interactions between elementary particles and are exemplified in Figure 1. The horizontal axis in these diagrams usually denotes space, while the vertical axis typically corresponds to time [15]. These diagrams make it possible to predict the probabilities of scattering events with high precision, which is essential for detecting new interactions and particles beyond the Standard Model. They can also determine whether certain scattering processes are possible or not [16]. Before giving a detailed description of these diagrams, we introduce some important concepts in graph theory.

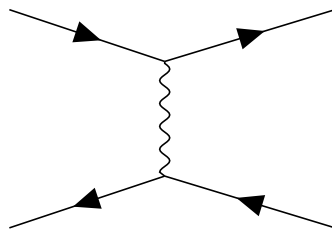


Figure 1: An example of a Feynman diagram.

1.4.1 Graph Theory and Set Theory

Graph theory is a branch of mathematics dealing with the study of structures consisting of vertices connected by edges [17]. This section defines some of its central concepts, such

as graphs, isomorphisms between graphs, subgraphs, cut vertices and cut edges. Before studying graph theory in more depth, we introduce a few key concepts in set theory.

1.4.1.1 Set Theory

Set theory is the study of well-defined collections of objects, known as sets. A set is a collection of elements in which the order does not matter [18]. For example, the sets $\{a, b, c\}$ and $\{b, c, a\}$ are considered equal. For two sets M and N , several standard set operations are defined as follows.

Definition 1.1 (Union). *The union $M \cup N$ is the set of all elements contained in M or N .*

Definition 1.2 (Intersection). *The intersection $M \cap N$ is the set of all elements contained in M and N .*

Definition 1.3 (Complement). *The complement $M \setminus N$ is the set of all elements contained in M , but not in N .*

For instance, if we consider the sets $M = \{1, 3, 5, 7\}$ and $N = \{2, 4, 6, 7\}$, then $M \cup N = \{1, 2, 3, 4, 5, 6, 7\}$, $M \cap N = \{7\}$ and $M \setminus N = \{1, 3, 5\}$. Note that $M \setminus N \neq N \setminus M$.

A set M is said to be a subset of another set N , denoted $M \subseteq N$, if every element in M is also an element in N .

1.4.1.2 Graph Theory

With the basics of set theory explained, we now define some key concepts in graph theory.

Definition 1.4 (Graph). *A graph G is a pair $G(V, E)$, where V is a set of objects called vertices, and E is a set of sets with two elements in V , which are known as edges.*

An example of a graph G where $V = \{a, b, c\}$ and $E = \{\{a, b\}, \{a, c\}, \{b, c\}\}$ is shown in Figure 2. The graph is denoted $G(V, E) = G(\{a, b, c\}, \{\{a, b\}, \{a, c\}, \{b, c\}\})$.

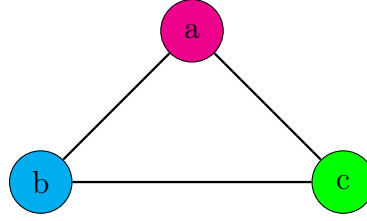


Figure 2: An example of a graph.

Definition 1.5 (Directed graph). *A directed graph G is a pair $G(V, E)$, where V is a set of vertices and E is a set of ordered pairs of elements of V , known as directed edges.*

An example of a directed graph G where $V = \{a, b, c\}$ and $E = \{(a, b), (a, c), (b, c)\}$ is shown in Figure 3. The graph is denoted $G(V, E) = G(\{a, b, c\}, \{(a, b), (a, c), (b, c)\})$. The difference between directed and undirected graphs lies in how their edges are represented. In directed graphs, edges are ordered pairs, while in undirected graphs, edges are sets.

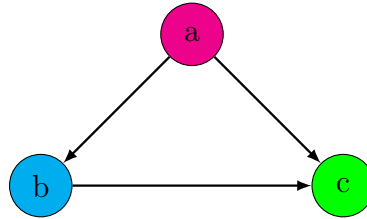


Figure 3: An example of a directed graph.

The sets V and E can be infinite. However, this paper only considers finite graphs, where V and E are finite sets. Furthermore, we assume that all graphs are undirected from this point onward.

Definition 1.6 (Loop). *A loop is an edge e that connects a vertex v to itself.*

An example of a graph containing a loop is shown in Figure 4. Since edges are sets, loops are distinct from ordinary edges in that they are represented as sets containing only a single element in V .

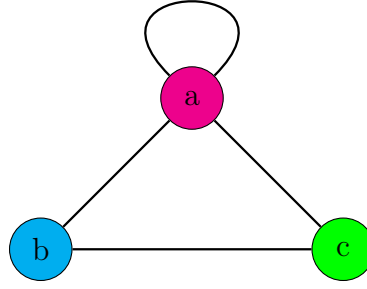


Figure 4: An example of a graph containing a loop.

Definition 1.7 (Multiple edges). *Multiple edges are edges that connect the same pair of vertices. Multiple edges consist of two elements in V and an index that distinguishes the edge from other edges between the same vertices.*

An example of multiple edges between two vertices is shown in Figure 5. The graph is denoted $G(\{a, b\}, \{\{a, b, 1\}, \{a, b, 2\}\})$.



Figure 5: An example of multiple edges between two vertices.

Definition 1.8 (Multigraph). *A multigraph is a graph G that may contain loops and multiple edges.*

An example of a multigraph is shown in Figure 6. Note that multigraphs do not necessarily contain multiple edges or loops.

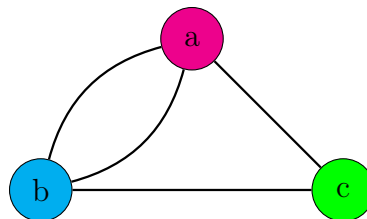


Figure 6: An example of a multigraph.

Henceforth, all graphs discussed in this paper will be assumed to be multigraphs.

Definition 1.9 (Subgraph). *A graph H is a subgraph of a graph G if the vertices of H are a subset of the vertices of G and the edges of H are a subset of the edges of G .*

Two examples of subgraphs of the graph in Figure 2 are shown in Figure 7.

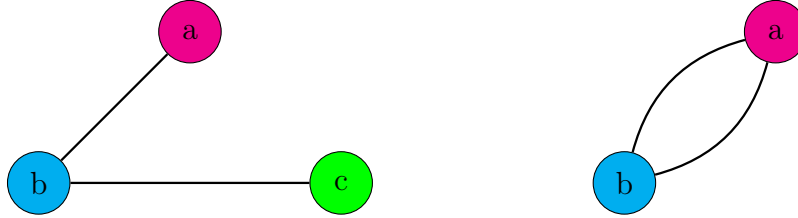


Figure 7: Two examples of subgraphs of the graph in Figure 2.

Definition 1.10 (Induced subgraph). *A subgraph $H(V', E')$ of a graph $G(V, E)$ is induced if and only if, for each pair of vertices $a, b \in V'$, there exist k edges connecting a and b in E' if and only if there exist exactly k edges connecting a and b in E .*

An example of an induced subgraph is shown in Figure 8. In the figure, the right graph is an induced subgraph of the left graph.

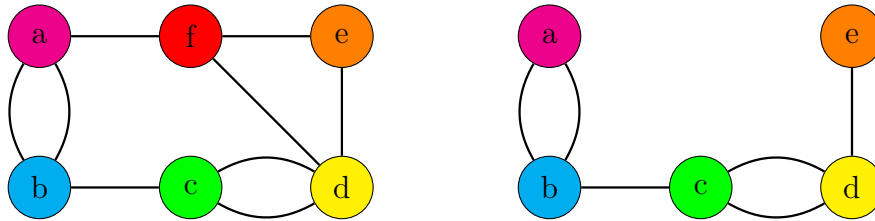


Figure 8: An example of an induced subgraph.

Definition 1.11 (Path). *A path is a sequence of edges connecting a sequence of vertices. In a path, all vertices and edges are distinct.*

An example of a path in a graph is shown in Figure 9. In the figure, a path connecting a and c is highlighted with orange color and bold edges.

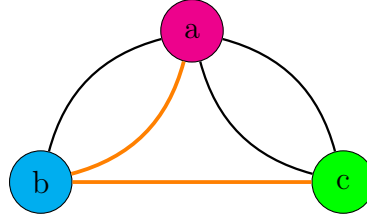


Figure 9: An example of a path in a graph.

Definition 1.12 (Connected graph). *A graph $G(V, E)$ is connected if and only if there exists a path between all pairs of vertices $u, v \in V$.*

An example of a connected graph and a disconnected graph is shown in Figure 10. In the figure, the left graph is connected and the right graph is disconnected.

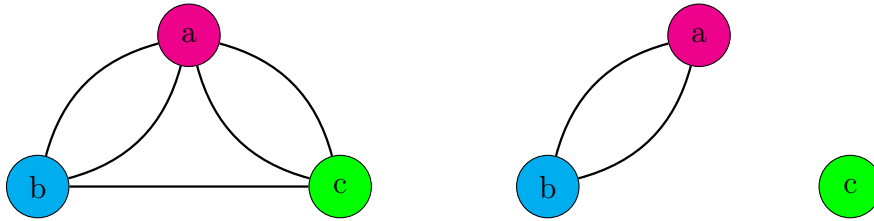


Figure 10: An example of a connected graph and a disconnected graph.

Definition 1.13 (Cut vertex and cut edge). *A cut vertex v in a graph $G(V, E)$ is a vertex, such that the induced subgraph $H(V', E')$ of $G(V, E)$, where $V' = V \setminus \{v\}$, is disconnected. Similarly, a cut edge $\{u, v\}$ in a graph $G(V, E)$ is an edge, such that the subgraph $H(V, E')$ of $G(V, E)$, where $E' = E \setminus \{e\}$, is disconnected.*

An example of a cut vertex and a cut edge in a graph is shown in Figure 11, where the vertex a is a cut vertex and the edge $\{a, b\}$ is a cut edge.

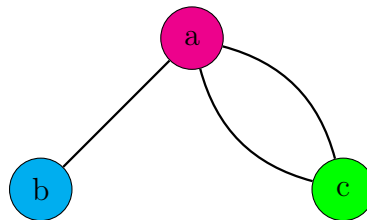


Figure 11: An example of a cut vertex and a cut edge in a graph.

Definition 1.14 (Degree of a vertex). *The degree of a vertex v in a graph $G(V, E)$, denoted $\deg(v)$, is the number of edges for which it is an endpoint. Loops count double towards the degree of a vertex.*

For example, consider $G(V, E) = G(\{a, b, c\}, \{\{a\}, \{a, c\}, \{a, b, 1\}, \{a, b, 2\}, \{b, c\}\})$, illustrated in Figure 12. Then, $\deg(a) = 5$, $\deg(b) = 3$ and $\deg(c) = 2$.

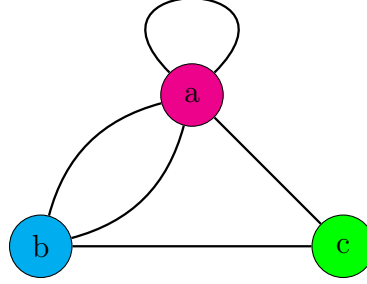


Figure 12: An example of a graph where $\deg(a) = 5$, $\deg(b) = 3$ and $\deg(c) = 2$.

Definition 1.15 (Graph Isomorphism). *The graphs $G(V, E)$ and $H(V', E')$ are isomorphic if and only if there exists a bijection $f : V \rightarrow V'$, such that an edge between vertices $v_1, v_2 \in V$ exists in E if and only if an edge between $f(v_1)$ and $f(v_2)$ exists in E' . A bijection f is a function that maps each element of a set to a unique element of another set, and vice versa.*

An example of two isomorphic graphs is shown in Figure 13. Determining graph isomorphism can be a complex task, so it is important to first compare simple properties that can quickly rule out isomorphism when designing isomorphism algorithms. Such properties include the number of vertices and edges, as well as the degree of each vertex.

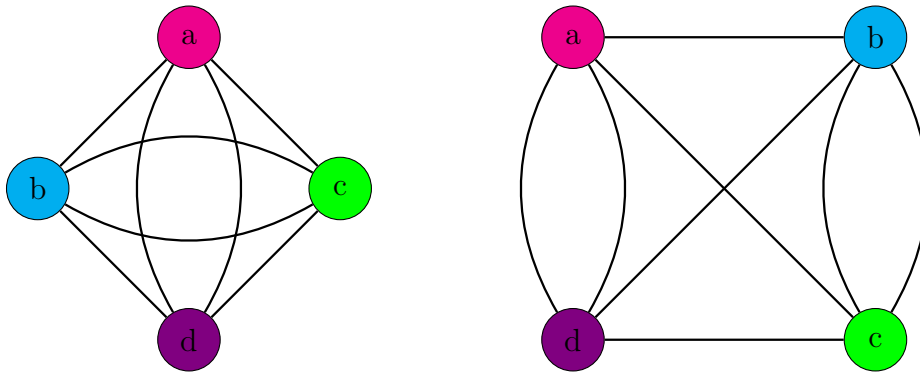


Figure 13: An example of two isomorphic graphs.

1.4.2 Feynman Diagrams Represented As Graphs

In Feynman diagrams, straight lines represent fermions, wavy lines represent bosons, and vertices represent interactions between particles [15]. If we disregard the type of particles

involved in the interactions shown by Feynman diagrams, these can be simplified to what we refer to as Feynman graphs. When transitioning from Feynman graphs to Feynman diagrams, you modify the edges, but you must also ensure that the created diagrams represent physically possible interactions.

Definition 1.16 (Feynman graph). *A Feynman graph of order k is a connected graph $G(V, E)$, such that every vertex $v \in V$, except for k vertices of degree one, called external vertices, is of degree k . The edges connected to the external vertices are called external legs.*

Feynman graphs of order k represent interactions in φ^k theory, and are referred to as φ^k Feynman graphs. An example of a φ^4 Feynman graph is shown in Figure 14. Throughout this paper, the external vertices are transparent. The number of interaction vertices in a Feynman graph refers to the number of vertices that are not external vertices. φ^4 Feynman graphs can be generated from so-called vacuum graphs of order four using an operator known as “Removing a vertex”. We first define a vacuum graph.

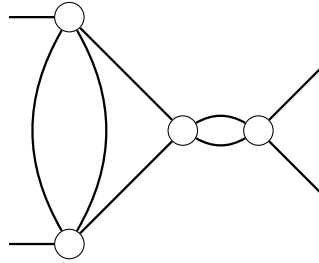


Figure 14: An example of a φ^4 Feynman graph.

Definition 1.17 (Vacuum graph). *A vacuum graph of order k is a connected graph free from loops, such that every vertex in the graph is of degree k .*

An example of a vacuum graph of order four is shown in Figure 15.

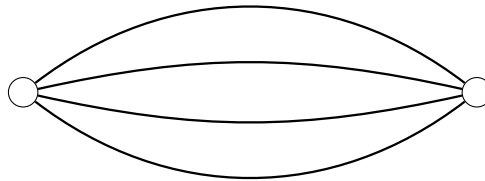


Figure 15: An example of a vacuum graph of order four.

We now define the operator “Removing a vertex”.

Definition 1.18 (Removing a vertex). *Let $G(V, E)$ be a vacuum graph of order four, v be a vertex in V and e_1, e_2, e_3, e_4 be the edges connected to v . Removing the vertex v from $G(V, E)$ generates the φ^4 Feynman graph $G(V', E')$, where $V' = V \setminus \{v\}$ and each edge connected to v has its endpoint switched from v to a distinct external vertex, transforming them into external legs.*

For example, if we use the operator Removing a vertex on a vertex in the vacuum graph shown in Figure 15, we get the φ^4 Feynman graph illustrated in Figure 16.

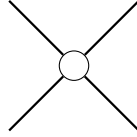


Figure 16: An example of a φ^4 Feynman graph created by the operator Removing a vertex.

1.4.3 Constraints on the Graphs

Using the definitions above, some φ^4 Feynman graphs describe particle interactions that are unnecessary to generate. These are graphs with cut edges, loops and what we refer to as generalized loops. Before explaining this in detail, we first define a generalized loop.

Definition 1.19 (Generalized loop). *A generalized loop is a connected induced subgraph $H(V', E')$ of a graph $G(V, E)$, such that H neither contains external legs nor has more than one vertex $v \in V'$ connected to vertices not in H . A generalized loop contains at least two vertices.*

An example of a generalized loop in a graph is shown in Figure 17. The generalized loop is denoted $H(\{a, b, c\}, \{\{a, b\}, \{a, c\}, \{b, c, 1\}, \{b, c, 2\}, \{b, c, 3\}\})$ and is highlighted with green color and bold edges.

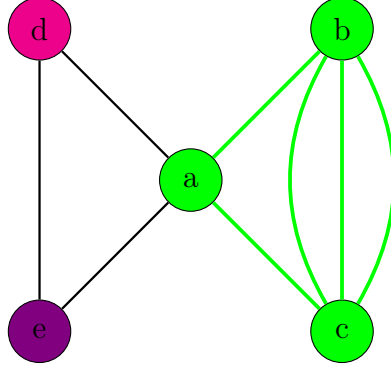


Figure 17: An example of a generalized loop in a graph.

We now introduce a theorem about generalized loops, crucial in the generation of φ^4 Feynman graphs.

Theorem 1.20. *Let $G(V, E)$ be a vacuum graph and v be a cut vertex in V . Then, removing any non-cut vertex yields a graph with a generalized loop.*

Proof. Since v is a cut vertex, there exist two connected induced subgraphs $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ of G , such that $V_1 \cap V_2 = \{v\}$ and such that G_1 and G_2 are generalized loops. The use of the operator Removing a vertex on a vertex in G_1 then creates a φ^4 Feynman graph with G_2 as a generalized loop, and vice versa. $\square$

φ^4 Feynman graphs with loops and generalized loops are discarded, as they do not represent physically possible particle interactions. Moreover, graphs with cut edges are excluded, as the removal of a cut edge separates a φ^4 Feynman graph into two subgraphs, each representing separate φ^4 particle interactions. The interactions visualized by the subgraphs can be analyzed independently, making it unnecessary to study them together.

Based on these constraints, we redefine a Feynman graph.

Definition 1.21 (Feynman graph, modified). *A Feynman graph of order k is a connected graph $G(V, E)$ free from cut edges, loops, and generalized loops, such that every vertex $v \in V$, except for k vertices of degree one, called external vertices, is of degree k . The edges connected to the external vertices are called external legs.*

Finally, we show that the operator Removing a vertex generates candidate φ^4 Feynman graphs. These satisfy all conditions for being φ^4 Feynman graphs, except that they may contain cut edges.

Theorem 1.22. *Removing a non-cut vertex from a vacuum graph of order four without a loop or generalized loop creates a candidate φ^4 Feynman graph.*

Proof. Let $G(V, E)$ be a vacuum graph of order four, v be a vertex of V and e_1, e_2, e_3, e_4 be the edges connected to v . After the vertex v is removed, the graph is connected because v is not a cut vertex, and moreover the degree of the vertices is preserved. This is because the edges that were connected to v have become external legs connected to external vertices. Since the vacuum graph does not contain any loop or generalized loop, the created graph will also be free from such structures. Thus, a candidate φ^4 Feynman graph has been generated. $\square$

Therefore, after applying the operator Removing a vertex on a vertex in a vacuum graph of order four, one must check for cut edges to ensure that the resulting graph is a valid φ^4 Feynman graph.

1.5 Aim of Study

Previous studies have counted the number of vacuum graphs of order four with a large number of vertices [19]. In contrast, φ^4 Feynman graphs have only been generated with up to six interaction vertices, and an algorithm to generate these has only been described in general terms [20]. The aim of this study is to develop a recursive algorithm that can generate all φ^4 Feynman graphs with a larger number of interaction vertices.

1.6 Research Question

How many non-isomorphic φ^4 Feynman graphs exist for each number of interaction vertices up to nine?

2 Method

The φ^4 Feynman graphs were generated using a recursive algorithm divided into two steps, as described in this section. The algorithm was implemented in the programming language

Python 3, and made use of the NetworkX library, which provides graph data structures and graph-theoretic algorithms [21]. The implementation can be found in Appendix A.

The first step involved recursively generating all vacuum graphs of order four, starting from graphs with one and two vertices, and continuing up to graphs with $k + 1$ vertices, where k is the number of interaction vertices in the φ^4 Feynman graphs being generated. In the second step, all φ^4 Feynman graphs with k interaction vertices were generated using the vacuum graphs with $k + 1$ vertices from the first step. In this paper, φ^4 Feynman graphs with up to nine interaction vertices were generated.

2.1 Generation of Vacuum Graphs

In the generation of vacuum graphs of order four, two operators were used: “Crossing a vertex” and “Tripling an edge”, which are defined as follows.

2.1.1 Defining the Operators

Definition 2.1 (Crossing a vertex). *Let $G(V, E)$ be a vacuum graph of order four, and e_1 and e_2 be two distinct edges in E , connecting vertex a to vertex b and vertex c to vertex d (where we potentially have $a = c$ or $b = d$). Crossing a vertex means removing e_1 and e_2 , and adding a vertex v as well as the edges $\{a, v\}$, $\{b, v\}$, $\{c, v\}$ and $\{d, v\}$.*

Crossing a vertex increases the number of vertices in the graph by one. An example of this operator applied on two edges in a vacuum graph of order four is shown in Figure 18. The edges $\{a, b\}$ and $\{c, d\}$ are highlighted in pink and are bold, the vertex v is highlighted in blue and the edges $\{a, v\}$, $\{b, v\}$, $\{c, v\}$ and $\{d, v\}$ are highlighted in blue and are bold.

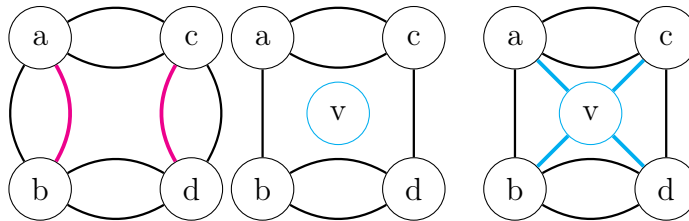


Figure 18: An example of the operator Crossing a vertex being performed.

Definition 2.2 (Tripling an edge). *Let $G(V, E)$ be a vacuum graph of order four and e an edge connecting vertices a and b . Tripling an edge means removing e and adding two vertices u and v , the edges $\{a, u\}$ and $\{b, v\}$, as well as three edges between u and v .*

Tripling an edge increases the number of vertices in the graph by two. An example of this operator applied on an edge in a vacuum graph of order four is shown in Figure 19. The edge $\{a, b\}$ is highlighted in pink and is bold, the vertices u and v are highlighted in blue and the edges $\{a, u\}$, $\{b, v\}$ as well as the three edges between u and v are highlighted in blue and are bold.

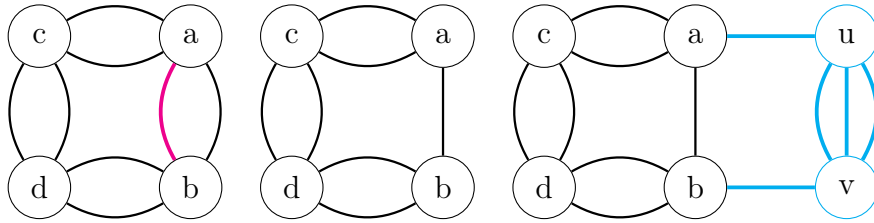


Figure 19: An example of the operator Tripling an edge being performed.

Lemma 2.3. *After the operator “Crossing a vertex” or “Tripling an edge” has been performed, the resulting graph is always a vacuum graph of order four.*

Proof. All added vertices are of degree four, and for each vertex connected to a removed edge, the number of new edges equals the number of edges removed. Consequently, each vertex in the graph remains of degree four. Moreover, the graph remains connected and free from loops, as the operators introduce paths between each pair of vertices previously connected by a removed edge, and the original graph contains no loops. It therefore follows from Definition 1.17 that the resulting graph is a vacuum graph of order four. $\square$

Thus, the operators generate vacuum graphs of order four. We now describe the process of generating such graphs and prove that all vacuum graphs of order four can be constructed using these operators.

2.1.2 Recursive Generation of Vacuum Graphs

The vacuum graphs of order four were generated recursively. First, the set V_1 , consisting of all vacuum graphs of order four with one vertex, and the set V_2 , consisting of all vacuum

graphs of order four with two vertices, were created. Since loops are not allowed, V_1 is empty and V_2 consists of a single graph, illustrated in Figure 20.

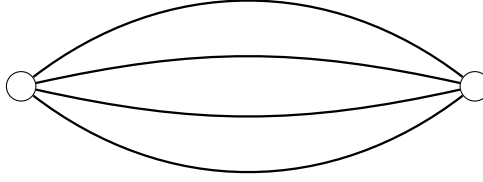


Figure 20: The one vacuum graph in V_2 .

Then, the empty sets $V_3, \dots, V_k$ were created recursively, where k is the number of vertices in the vacuum graphs to be generated (for $k \geq 3$). To create these sets, all graphs in the previous set were iterated through. The operator Crossing a vertex was applied to each pair of edges to produce new vacuum graphs of order four.

When the current set was empty, the created vacuum graph was added directly. Otherwise, isomorphism checks were performed against all existing graphs in the set. If an isomorphism was detected, the graph was discarded. If not, the graph was added to the set.

Next, all graphs in the set of graphs with two fewer vertices than the set to be created were iterated through, along with all their edges. The operator Tripling an edge was performed on each edge, and the resulting graphs were checked for isomorphisms and added to the set in the same way as the graphs created with the operator Crossing a vertex. Once all graphs in V_{k-1} and V_{k-2} had been processed, all vacuum graphs of order four with k vertices had been generated, as we will now show.

2.1.3 Proof that the Recursion Generates All Vacuum Graphs

We now demonstrate that any vacuum graph of order four with k vertices (for $k \geq 3$) can be generated from the sets of vacuum graphs with one and two fewer vertices using the operators Crossing a vertex and Tripling an edge. To prove this, we show that for every vacuum graph G of order four with more than two vertices, there exists a vacuum graph H of order four, where one of the above operators can be performed on one or two edges in H to obtain G .

Lemma 2.4 (Reversing Crossing a vertex). *Let $G(V, E)$ be a vacuum graph of order four, in which there exists a vertex v connected to:*

- *Four distinct vertices,*
- *Three distinct vertices, or*
- *Two distinct vertices, connected to v with two edges each.*

In these cases, there exists some vacuum graph $H(V', E')$ of order four such that performing the operator Crossing a vertex on some pair of edges in H yields G .

A vacuum graph of order four with vertices satisfying the conditions of Lemma 2.4 is shown in Figure 21.

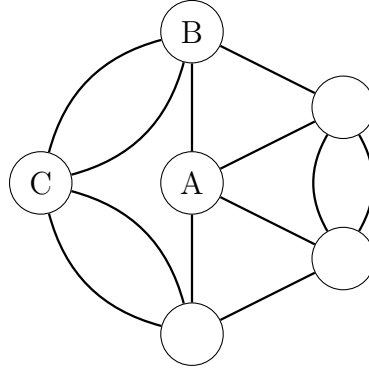


Figure 21: A vacuum graph with vertices satisfying the conditions in Lemma 2.4.

Proof. We prove all three cases individually.

Case 1: There exists a vertex v connected to the vertices a, b, c and d . Consider the graph $G'(V', E')$, with $V' = V \setminus \{v\}$ and $E' = E \setminus \{\{a, v\}, \{b, v\}, \{c, v\}, \{d, v\}\} \cup \{\{a, b, 2\}, \{c, d, 2\}\}$. Applying Crossing a vertex on $\{a, b, 2\}$ and $\{c, d, 2\}$ in this graph creates G .

To exemplify this, we consider the graph in Figure 22, where the operator Crossing a vertex creates the graph in Figure 21 when the edges $\{a, b, 2\}$ and $\{c, d, 2\}$ are removed, and the vertex A as well as the edges $\{A, a\}$, $\{A, b\}$, $\{A, c\}$, and $\{A, d\}$ are added.

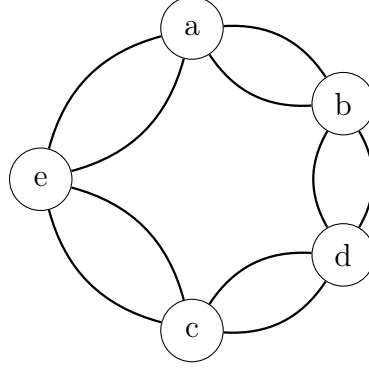


Figure 22: A graph that, when using Crossing a vertex, yields the graph in Figure 21.

Case 2: There exists a vertex v connected to vertices a , b and c . Since $\deg(v) = 4$, there must exist two edges connecting v to one of the other vertices. Without loss of generality, we assume that two edges connecting a and v exist. Consider the graph $G'(V', E')$ where $V' = V \setminus \{v\}$ and $E' = E \setminus \{\{a, v, 1\}, \{a, v, 2\}, \{b, v\}, \{c, v\}\} \cup \{\{a, b\}, \{a, c\}\}$. Applying Crossing a vertex on $\{a, b\}$ and $\{a, c\}$ in this graph generates G .

To exemplify this, we consider the graph in Figure 23, where the operator Crossing a vertex creates the graph in Figure 21 when the edges $\{a, b\}$ and $\{a, c\}$ are removed, and the vertex B as well as the edges $\{B, a, 1\}$, $\{B, a, 2\}$, $\{B, b\}$, and $\{B, c\}$ are added.

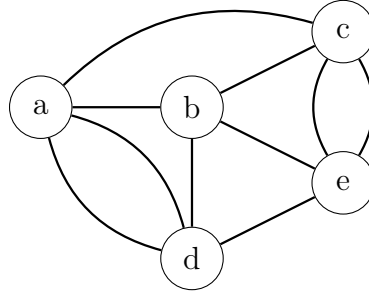


Figure 23: A graph that, when using Crossing a vertex, yields the graph in Figure 21.

Case 3: There exists a vertex v connected to vertices a and b , with two edges connecting v and a and two edges connecting v and b . Consider the graph $G'(V', E')$ where $V' = V \setminus \{v\}$ and $E' = E \setminus \{\{a, v, 1\}, \{a, v, 2\}, \{b, v, 1\}, \{b, v, 2\}\} \cup \{\{a, b, 1\}, \{a, b, 2\}\}$. Applying Crossing a vertex on $\{a, b, 1\}$ and $\{a, b, 2\}$ in this graph generates G .

To exemplify this, we consider the graph in Figure 24, where the operator Crossing a vertex creates the graph in Figure 21 when the edges $\{b, c, 1\}$ and $\{b, c, 2\}$ are removed, and the vertex C as well as the edges $\{C, b, 1\}$, $\{C, b, 2\}$, $\{C, c, 1\}$ and $\{C, c, 2\}$ are added.

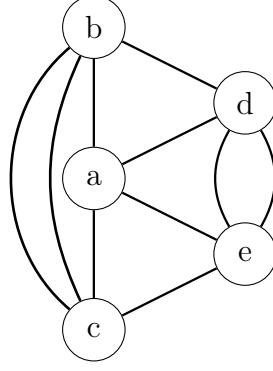


Figure 24: A graph that, when using Crossing a vertex, yields the graph in Figure 21.

Thus, there exists at least one graph G' that satisfies the requirement. $\square$

Lemma 2.5 (Reversing Tripling an edge). *Let $G(V, E)$ be a vacuum graph of order four and a and b be vertices in V , connected by exactly three edges, and such that no vertex connected to both a and b exists. Then, there exists a vacuum graph $G'(V', E')$ of order four, such that applying the operator Tripling an edge on an edge in G' yields G .*

Proof. Since $\deg(a) = \deg(b) = 4$, the vertices c and d exist, such that a and c , and b and d , respectively, are connected. Then, applying the operator Tripling an edge on an edge in G' , where $V' = V \setminus \{a, b\}$ and $E' = E \setminus \{\{a, c\}, \{b, d\}, \{a, b, 1\}, \{a, b, 2\}, \{a, b, 3\}\} \cup \{\{c, d\}\}$ yields the graph G .

To exemplify this, we consider the left graph in Figure 25, which, using the operator Tripling an edge, creates the right graph when the edge $\{a, b, 1\}$ is removed, and the vertices u and v as well as the edges $\{a, u\}$, $\{b, v\}$, $\{u, v, 1\}$, $\{u, v, 2\}$, $\{u, v, 3\}$ are added. The edge $\{a, b\}$ is highlighted in blue and is bold, the vertices u and v are highlighted in red and the edges $\{a, u\}$, $\{b, v\}$ as well as the three edges between u and v are highlighted in red and are bold.

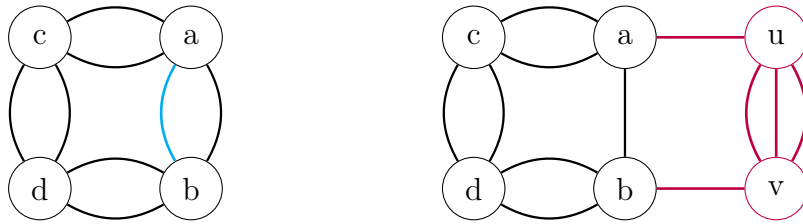


Figure 25: A graph that, when using Tripling an edge, yields a vacuum graph.

$\square$

We now show that the operators Crossing a vertex and Tripling an edge can be used to recursively generate all vacuum graphs of order four.

Theorem 2.6. *Let $G(V, E)$ be a graph in V_k , where $k \geq 3$. Then, there exists a graph in V_{k-1} , such that performing Crossing a vertex on a pair of edges in the graph creates G , or a graph in V_{k-2} , such that applying Tripling an edge on an edge in the graph yields G .*

Proof. Let a and b be two vertices in $G(V, E)$. First, assume that exactly three edges connecting a and b exist. Since $k \geq 3$, there must exist a vertex c connected to a , and a vertex d connected to b . If $c \neq d$, there exists a graph that generates G with the operator Tripling an edge, according to Lemma 2.5. If $c = d$, then c is connected to a and b with one edge each. Since $\deg(c) = 4$, c must be connected to three or four distinct vertices. By Lemma 2.4, there then exists a graph from which G is created using the operator Crossing a vertex. Thus, a desired graph always exists in V_{k-1} or V_{k-2} in this case. Otherwise, consider a vertex $v \in V$. Since no vertex is connected to v by exactly three edges and $\deg(v) = 4$, the following cases arise:

- v is connected to four distinct vertices,
- v is connected to three distinct vertices, and
- v is connected to two distinct vertices, which are connected to v by two edges each.

In all cases, by Lemma 2.4, a graph such that when applying Crossing a vertex, creates G exists. Thus, a vacuum graph of order four that satisfies the conditions always exists. □

We now introduce a corollary about the graphs that the sets V_k contain.

Corollary 2.7. *Let V_{k-1} and V_{k-2} be the sets of the vacuum graphs of order four with $k-1$ and $k-2$ vertices, respectively (for $k \geq 3$). Then, V_k consists precisely of all graphs generated by applying Crossing a vertex to pairs of edges in graphs in V_{k-1} and Tripling an edge to edges in graphs in V_{k-2} .*

The corollary follows directly from Theorem 2.6 and shows that a recursive algorithm to generate the sets $V_3, \dots, V_k$ for any $k > 2$ can be constructed using the operators Crossing a vertex and Tripling an edge. Since V_1 and V_2 are known, a recursive algorithm starting from the sets V_1 and V_2 can be constructed to generate V_k for all $k \geq 3$.

2.2 Generation of Feynman Graphs

Once all the vacuum graphs of order four with $k + 1$ vertices, where k is the number of vertices in the desired φ^4 Feynman graphs, had been generated, the φ^4 Feynman graphs with k vertices were obtained using the operator Removing a vertex. We now describe the method for generating φ^4 Feynman graphs, and then show that the operator can generate all such Feynman graphs.

2.2.1 Using the Operator for Graph Generation

First, the empty set F_k was created, where k is the number of vertices in the φ^4 Feynman graphs to be generated. To construct the graphs, the algorithm iterated through all the graphs in V_{k+1} checking for cut vertices.

Graphs containing cut vertices were discarded. For the remaining graphs, each vertex was removed in turn using the operator Removing a vertex, producing a candidate φ^4 Feynman graph, as guaranteed by Theorem 1.22. The resulting graph was then checked for cut edges, and if any were found the graph was discarded.

If no cut edges were present, isomorphism checks were performed against all previously generated graphs from the same vacuum graph. Non-isomorphic graphs were added to F_k .

2.2.2 Proof That the Algorithm Generates All Feynman Graphs

We now show that all φ^4 Feynman graphs with k vertices can be created using the set V_{k+1} .

Theorem 2.8. *For every φ^4 Feynman graph, there exists a unique vacuum graph $G(V, E)$ of order four and at least one vertex $v \in V$, such that removing v from the vacuum graph yields the desired Feynman graph.*

Proof. Consider a graph where all the external legs of a φ^4 Feynman graph, instead of connecting to external vertices, connect to a single vertex. This graph is connected and without loops, and all of its vertices have order four, making it a vacuum graph of order four. Thus, removing this vertex from the vacuum graph generates the desired φ^4 Feynman graph. Furthermore, this vacuum graph is unique because the operator Removing a vertex creates a unique graph. $\square$

Thus, the method for generating φ^4 -Feynman graphs works.

2.3 Performance Analysis of the Algorithm

The total runtime of the algorithm and its key components was measured during the generation of φ^4 Feynman graphs with up to nine interaction vertices. More specifically, the time consumption of the following parts was measured:

- Generating all φ^4 Feynman graphs,
- Generating all vacuum graphs of order four,
- Checking for cut edges,
- Checking for cut vertices,
- Checking for isomorphisms, and
- Performing Crossing a vertex and Tripling an edge.

In addition, the runtime of the two components of the isomorphism check was measured separately. The first one compares the number of double edges, triple edges and cycles, which are paths connecting vertices to themselves, of length three. The second one uses built-in functions in NetworkX. It is important to note that the total execution time does not necessarily equal the sum of the individual component run-times, as some parts of the algorithm are not part of these sub-algorithms. The number of graphs identified as isomorphic to others during the isomorphism checks was also recorded.

3 Results

All generated φ^4 Feynman graphs with up to four interaction vertices are shown in Figures 26–29, while the number of generated graphs with up to nine interaction vertices is listed in Table 1. The measured run-times for the algorithm’s main components and the number of isomorphisms detected during the isomorphism checks are listed in Appendix B.

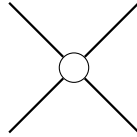


Figure 26: The only φ^4 -Feynman graph with one interaction vertex.

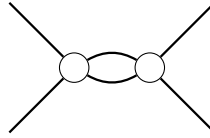


Figure 27: The only φ^4 -Feynman graph with two interaction vertices.

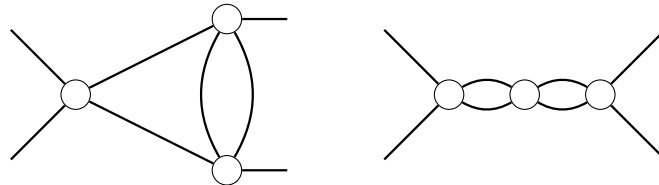


Figure 28: All φ^4 Feynman graphs with three interaction vertices.

Table 1: The total number of φ^4 Feynman graphs for up to nine interaction vertices.

Number of Interaction Vertices	Number of φ^4 Feynman Graphs
1	1
2	1
3	2
4	8
5	26
6	124
7	627
8	3794
9	25306

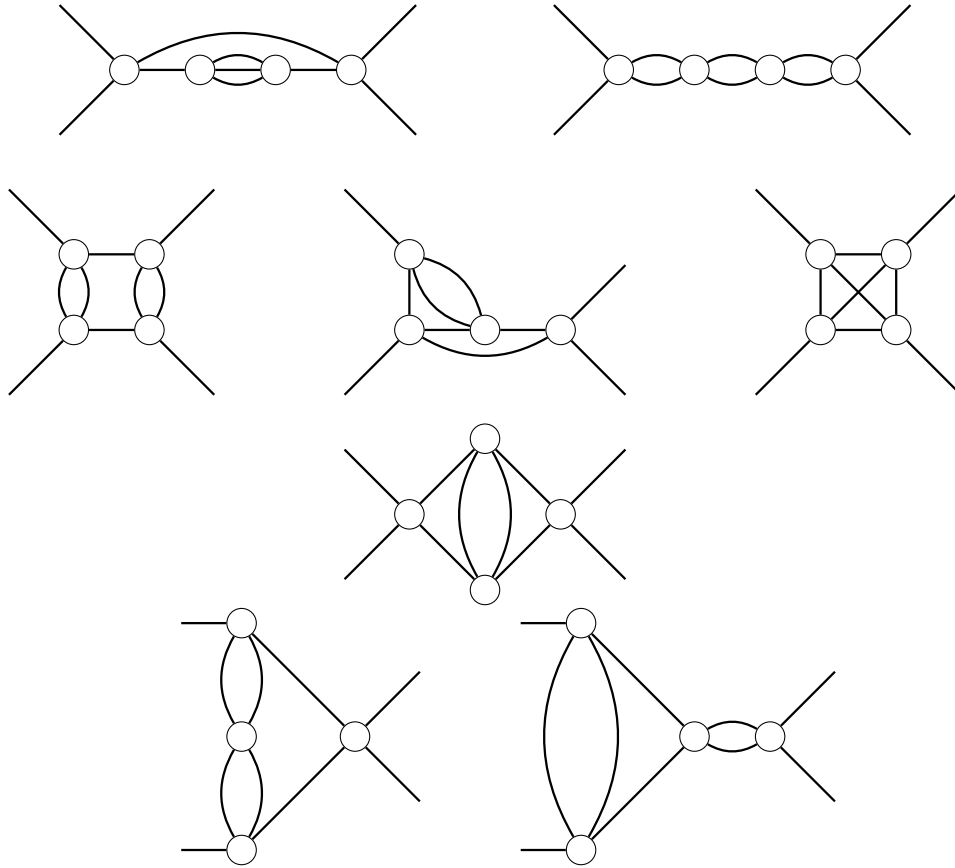


Figure 29: All φ^4 Feynman graphs with four interaction vertices.

4 Discussion

The algorithm developed to generate the φ^4 Feynman graphs successfully produced valid, non-isomorphic graphs, consistent with those described by Kleinert and Schulte-Frohlinde [20]. The open-source nature of the code is vital for the research community. By making it publicly available, researchers can easily replicate and build upon the work, speeding up progress. Access to these φ^4 Feynman graphs, essential for calculating scattering amplitudes, enables further study of physical phenomena, advancing theoretical research.

However, there are potential improvements to the code that could enhance its performance. Before discussing these, the algorithm's time complexity is briefly addressed.

4.1 The Algorithm's Time Complexity

An estimate of the algorithm's time complexity can be made by analyzing the time complexities of its different sub-algorithms, which are based on NetworkX [21], as shown in Table 2. In the table, v and e represent the number of vertices and edges in the graphs, respectively, while $f(v)$ denotes the growth rate of the number of vacuum graphs of degree four. Some sub-algorithms with negligible time complexity are excluded from the table, although they may play a significant role when generating graphs with a small number of interaction vertices.

Table 2: Time complexities for different sub-algorithms.

Sub-algorithm	Time Complexity
Check for cut edges	$O(v + e)$
Check for cut vertices	$O(v + e)$
Slow isomorphism check	$O(v! \cdot v \cdot f(v))$
Fast isomorphism check	$O(v \cdot f(v))$

Before the total time complexity can be calculated, we need to estimate $f(v)$, which requires more extensive research. As a result, providing an accurate estimate of the time complexity for the entire algorithm is challenging. However, preliminary results along with previous research [19] suggest that $f(v)$ is approximately $O(v!)$.

4.2 Further Studies

As shown in Appendix B, the isomorphism checks account for the majority of the algorithm's execution time. Therefore, developing a more efficient algorithm for isomorphism checks could significantly reduce the overall runtime. Additionally, utilizing parallel processing to run the code on multiple cores could further improve performance. For instance, parallelizing isomorphism checks and applying the operator Removing a vertex on vertices in vacuum graphs could lead to substantial efficiency gains, as more operations could be performed simultaneously.

Beyond improving the algorithm's efficiency, future research could explore the development of algorithms that generate other types of Feynman graphs with different numbers

of vertices. Such developments could expand the algorithm's applicability, enabling the study of other particle interactions and systems within the same universality class as these graphs. Another possible improvement is to generate Feynman graphs with labeled edges, representing different particle types. This would require an additional validation step, ensuring that the generated graphs only represent physically possible interactions.

4.3 Conclusion

The number of non-isomorphic φ^4 Feynman graphs that exist for each number of interaction vertices up to nine are one, one, two, eight, 26, 124, 627, 3794 and 25306.

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A Code for Generating φ^4 Feynman Graphs

The implementation of the algorithm can be found here:

<https://github.com/klarabragesjobjorke/feynmangraphs>.

B Additional Statistics

Tables 3–6 display the runtime for the key sub-algorithms. Additionally, Table 7 presents the number of isomorphisms detected by the different parts of the isomorphism check.

Table 3: Runtime for the different parts of the algorithm.

Number of Vertices	Total Time	Time for Creation of Vacuum Graphs	Time for Creation of φ^4 -Feynman Graphs
1	$5.8 \cdot 10^{-3}$ s	$0.0 \cdot 10^{-4}$ s	$5.7 \cdot 10^{-3}$ s
2	$9.6 \cdot 10^{-3}$ s	$4.6 \cdot 10^{-3}$ s	$4.8 \cdot 10^{-3}$ s
3	$2.4 \cdot 10^{-3}$ s	$1.5 \cdot 10^{-3}$ s	$8.3 \cdot 10^{-3}$ s
4	$7.2 \cdot 10^{-3}$ s	$6.1 \cdot 10^{-3}$ s	$1.1 \cdot 10^{-3}$ s
5	$2.3 \cdot 10^{-3}$ s	$2.0 \cdot 10^{-3}$ s	$2.9 \cdot 10^{-3}$ s
6	2.2 s	2.1 s	$1.1 \cdot 10^{-3}$ s
7	2.8 s	2.7 s	$6.2 \cdot 10^{-3}$ s
8	$6.9 \cdot 10^2$ s	$6.8 \cdot 10^2$ s	3.8 s
9	$2.2 \cdot 10^4$ s	$2.2 \cdot 10^4$ s	$3.0 \cdot 10^1$ s

Table 4: Time taken to find cut edges and cut vertices.

Number of Vertices	Time to Find Cut Edges	Time to Find Cut Vertices
1	$2.2 \cdot 10^{-3}$ s	$1.1 \cdot 10^{-3}$ s
2	$1.9 \cdot 10^{-3}$ s	$9.0 \cdot 10^{-4}$ s
3	$3.1 \cdot 10^{-3}$ s	$7.8 \cdot 10^{-4}$ s
4	$3.4 \cdot 10^{-3}$ s	$4.8 \cdot 10^{-4}$ s
5	$9.5 \cdot 10^{-3}$ s	$5.5 \cdot 10^{-4}$ s
6	$3.2 \cdot 10^{-2}$ s	$1.0 \cdot 10^{-3}$ s
7	$1.7 \cdot 10^{-1}$ s	$3.4 \cdot 10^{-3}$ s
8	$8.1 \cdot 10^{-1}$ s	$1.4 \cdot 10^{-2}$ s
9	5.8 s	$8.3 \cdot 10^{-2}$ s

Table 5: Time taken to perform the operators.

Number of Vertices	Time to Perform Operators
1	$2.1 \cdot 10^{-4}$ s
2	$7.3 \cdot 10^{-4}$ s
3	$1.9 \cdot 10^{-3}$ s
4	$4.0 \cdot 10^{-3}$ s
5	$1.0 \cdot 10^{-2}$ s
6	$3.3 \cdot 10^{-2}$ s
7	$1.5 \cdot 10^{-1}$ s
8	$8.6 \cdot 10^{-1}$ s
9	5.1 s

Table 6: Time taken for different parts of the isomorphism check.

Number of Vertices	Total Time for Isomorphism Checks	Time for the First Part	Time for the Second Part
1	$1.5 \cdot 10^{-3}$ s	$0.0 \cdot 10^{-4}$ s	$0.0 \cdot 10^{-4}$ s
2	$4.3 \cdot 10^{-3}$ s	$0.0 \cdot 10^{-4}$ s	$0.0 \cdot 10^{-4}$ s
3	$2.4 \cdot 10^{-2}$ s	$5.5 \cdot 10^{-4}$ s	$0.0 \cdot 10^{-4}$ s
4	$5.2 \cdot 10^{-2}$ s	$5.1 \cdot 10^{-3}$ s	$1.7 \cdot 10^{-4}$ s
5	$1.8 \cdot 10^{-1}$ s	$3.4 \cdot 10^{-2}$ s	$2.0 \cdot 10^{-2}$ s
6	2.0 s	$4.7 \cdot 10^{-1}$ s	$8.2 \cdot 10^{-1}$ s
7	$2.7 \cdot 10^1$ s	7.7 s	$1.4 \cdot 10^1$ s
8	$6.8 \cdot 10^2$ s	$2.0 \cdot 10^2$ s	$3.7 \cdot 10^2$ s
9	$2.2 \cdot 10^4$ s	$7.0 \cdot 10^3$ s	$1.1 \cdot 10^4$ s

Table 7: Number of isomorphisms found by the isomorphism checks.

Number of Vertices	Isomorphisms Found by the First Part	Isomorphisms Found by the Second Part
1	0	0
2	0	0
3	32	0
4	422	2
5	4245	172
6	54283	5506
7	785803	88463
8	17457101	2117305
9	473037549	57291314

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